

EM for a Mixture of Two 1D Gaussians

We now derive the Expectation Maximization (EM) algorithm for a mixture of two one-dimensional Gaussian distributions. This is one of the most important and widely used instances of EM, and it clearly illustrates how the general framework translates into concrete parameter updates.

Suppose we are given observations

$$X = \{x_1, x_2, \dots, x_n\}, \quad x_i \in \mathbb{R}.$$

We assume that each data point is generated from one of two Gaussian distributions:

$$\mathcal{N}(\mu_1, \sigma_1^2), \quad \mathcal{N}(\mu_2, \sigma_2^2),$$

with mixing probabilities

$$\pi \quad \text{and} \quad 1 - \pi, \quad \text{where } 0 < \pi < 1.$$

Thus, the marginal likelihood of a data point is

$$p(x_i \mid \theta) = \pi \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + (1 - \pi) \mathcal{N}(x_i \mid \mu_2, \sigma_2^2),$$

where the parameter set is

$$\theta = \{\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2\}.$$

To express the model more conveniently, we introduce a latent variable $z_i \in \{0, 1\}$:

$$z_i = \begin{cases} 1 & \text{if data point } x_i \text{ comes from Gaussian 1,} \\ 0 & \text{if it comes from Gaussian 2.} \end{cases}$$

The complete-data likelihood then factorizes as

$$p(x_i, z_i \mid \theta) = [\pi \mathcal{N}(x_i \mid \mu_1, \sigma_1^2)]^{z_i} [(1 - \pi) \mathcal{N}(x_i \mid \mu_2, \sigma_2^2)]^{1-z_i}.$$

Taking logarithms and summing over all data points gives the complete-data log-likelihood:

$$\log p(X, Z \mid \theta) = \sum_{i=1}^n \left[z_i \log \pi + (1 - z_i) \log(1 - \pi) + z_i \log \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + (1 - z_i) \log \mathcal{N}(x_i \mid \mu_2, \sigma_2^2) \right].$$

If the z_i were known, we could directly maximize this expression. However, since they are hidden, we instead work with their expectations.

In the E-step, we compute the posterior probability that each point belongs to Gaussian 1:

$$\gamma_i := P(z_i = 1 \mid x_i, \theta^{(t)}).$$

Using Bayes' rule, this becomes

$$\gamma_i = \frac{\pi^{(t)} \mathcal{N}(x_i | \mu_1^{(t)}, \sigma_1^{2(t)})}{\pi^{(t)} \mathcal{N}(x_i | \mu_1^{(t)}, \sigma_1^{2(t)}) + (1 - \pi^{(t)}) \mathcal{N}(x_i | \mu_2^{(t)}, \sigma_2^{2(t)})}.$$

These quantities γ_i are often called *responsibilities*, since they measure how much Gaussian 1 is responsible for generating x_i . Naturally, $1 - \gamma_i$ represents the responsibility of Gaussian 2.

We now explain in detail how the expression for the Q -function arises in the Gaussian mixture model, and why the hidden variables z_i get replaced by their expectations γ_i .

Recall that we introduced latent variables

$$z_i \in \{0, 1\},$$

where $z_i = 1$ indicates that x_i was generated from Gaussian 1, and $z_i = 0$ indicates Gaussian 2.

Using this representation, the complete-data likelihood for a single data point can be written compactly as

$$p(x_i, z_i | \theta) = [\pi \mathcal{N}(x_i | \mu_1, \sigma_1^2)]^{z_i} [(1 - \pi) \mathcal{N}(x_i | \mu_2, \sigma_2^2)]^{1 - z_i}.$$

Why this form works. Because z_i is either 0 or 1:

- If $z_i = 1$, the expression reduces to $\pi \mathcal{N}(x_i | \mu_1, \sigma_1^2)$
- If $z_i = 0$, it reduces to $(1 - \pi) \mathcal{N}(x_i | \mu_2, \sigma_2^2)$

Thus, this single expression neatly encodes both cases.

Taking logarithms and summing over all data points, the complete-data log-likelihood becomes

$$\log p(X, Z | \theta) = \sum_{i=1}^n \log p(x_i, z_i | \theta).$$

Substituting the expression above and using $\log(a^b) = b \log a$, we obtain

$$\log p(X, Z | \theta) = \sum_{i=1}^n \left[z_i \log \pi + (1 - z_i) \log(1 - \pi) + z_i \log \mathcal{N}(x_i | \mu_1, \sigma_1^2) + (1 - z_i) \log \mathcal{N}(x_i | \mu_2, \sigma_2^2) \right].$$

Interpretation. Each data point contributes to exactly one component, but the indicator z_i allows us to write everything in a unified algebraic form.

At this stage, if the z_i were known, we could directly maximize this expression with respect to θ . However, since z_i is hidden, we instead work with its *expected value* under the current model.

In the E-step, we compute

$$\gamma_i := \mathbb{E}[z_i | x_i, \theta^{(t)}] = P(z_i = 1 | x_i, \theta^{(t)}).$$

Since z_i is a Bernoulli random variable, we also have

$$\mathbb{E}[1 - z_i] = 1 - \gamma_i.$$

Now, the Q -function is defined as the expectation of the complete-data log-likelihood:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z|X, \theta^{(t)}} [\log p(X, Z \mid \theta)].$$

Substituting the expression for $\log p(X, Z \mid \theta)$, we get

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \mathbb{E} \left[z_i \log \pi + (1 - z_i) \log(1 - \pi) + z_i \log \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + (1 - z_i) \log \mathcal{N}(x_i \mid \mu_2, \sigma_2^2) \right].$$

Using linearity of expectation, we can move the expectation inside:

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \left[\mathbb{E}[z_i] \log \pi + \mathbb{E}[1 - z_i] \log(1 - \pi) + \mathbb{E}[z_i] \log \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + \mathbb{E}[1 - z_i] \log \mathcal{N}(x_i \mid \mu_2, \sigma_2^2) \right].$$

Substituting $\mathbb{E}[z_i] = \gamma_i$ gives

$$Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^n \left[\gamma_i \log \pi + (1 - \gamma_i) \log(1 - \pi) + \gamma_i \log \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + (1 - \gamma_i) \log \mathcal{N}(x_i \mid \mu_2, \sigma_2^2) \right].$$

Key insight. We have not arbitrarily replaced z_i by γ_i . Rather, we have:

- Taken the expectation of the complete-data log-likelihood
- Used the fact that $\mathbb{E}[z_i] = \gamma_i$
- Used linearity of expectation to distribute over all terms

Interpretation. Each data point now contributes to both Gaussian components:

- With weight γ_i to component 1
- With weight $1 - \gamma_i$ to component 2

Thus, the hard assignment in the complete-data case is replaced by a *soft assignment*, which is the defining feature of EM. We now maximize this with respect to each parameter.

Update for π . We maximize

$$\sum_{i=1}^n [\gamma_i \log \pi + (1 - \gamma_i) \log(1 - \pi)].$$

Taking derivative and setting to zero:

$$\frac{\sum_i \gamma_i}{\pi} - \frac{\sum_i (1 - \gamma_i)}{1 - \pi} = 0.$$

Solving gives:

$$\pi^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \gamma_i.$$

Update for μ_1 . We maximize

$$\sum_{i=1}^n \gamma_i \log \mathcal{N}(x_i \mid \mu_1, \sigma_1^2).$$

Ignoring constants, this reduces to minimizing

$$\sum_i \gamma_i (x_i - \mu_1)^2.$$

Taking derivative:

$$\sum_i \gamma_i (x_i - \mu_1) = 0,$$

which gives

$$\mu_1^{(t+1)} = \frac{\sum_i \gamma_i x_i}{\sum_i \gamma_i}.$$

Similarly,

$$\mu_2^{(t+1)} = \frac{\sum_i (1 - \gamma_i) x_i}{\sum_i (1 - \gamma_i)}.$$

Update for σ_1^2 . We maximize

$$\sum_i \gamma_i \log \mathcal{N}(x_i | \mu_1, \sigma_1^2).$$

This leads to:

$$\sigma_1^{2(t+1)} = \frac{\sum_i \gamma_i (x_i - \mu_1^{(t+1)})^2}{\sum_i \gamma_i}.$$

Similarly,

$$\sigma_2^{2(t+1)} = \frac{\sum_i (1 - \gamma_i) (x_i - \mu_2^{(t+1)})^2}{\sum_i (1 - \gamma_i)}.$$

Putting everything together, the EM updates for a two-component Gaussian mixture are:

$$\begin{aligned} \gamma_i &= \frac{\pi \mathcal{N}(x_i | \mu_1, \sigma_1^2)}{\pi \mathcal{N}(x_i | \mu_1, \sigma_1^2) + (1 - \pi) \mathcal{N}(x_i | \mu_2, \sigma_2^2)} \\ \pi &= \frac{1}{n} \sum_i \gamma_i \\ \mu_1 &= \frac{\sum_i \gamma_i x_i}{\sum_i \gamma_i}, \quad \mu_2 = \frac{\sum_i (1 - \gamma_i) x_i}{\sum_i (1 - \gamma_i)} \\ \sigma_1^2 &= \frac{\sum_i \gamma_i (x_i - \mu_1)^2}{\sum_i \gamma_i}, \quad \sigma_2^2 = \frac{\sum_i (1 - \gamma_i) (x_i - \mu_2)^2}{\sum_i (1 - \gamma_i)} \end{aligned}$$

These updates have a very natural interpretation. Each data point contributes to both Gaussian components, but with different weights. The means are weighted averages, the variances are weighted variances, and the mixing coefficient is simply the average responsibility.

Thus, just as in the coin example, EM alternates between estimating hidden assignments (E-step) and updating parameters using those estimates (M-step), gradually refining both until convergence.